

Analysis & Modeling — Technical Reference

NHL Player Performance Model

Model Overview

The model uses **LightGBM** (Light Gradient Boosted Machine), a tree-based ensemble algorithm suited for tabular sports data. Two separate model pipelines are trained — one for forwards and one for defensemen — each producing three targets: **Points/Game**, **Goals/Game**, and **Game Score/Game**.

Residual Modeling

Rather than predicting raw stat values, each model predicts the **residual** — the deviation from a leakage-safe baseline computed from the player's own prior-season history (3-year rolling average, career average, and previous-season value, in that priority order). This means the model focuses its learning capacity on what makes a player outperform or underperform their own historical baseline, rather than learning that good players score more, which it already knows.

Feature Engineering

Features fall into four groups:

1. Player Skill Features

Derived from MoneyPuck shot tracking data. Includes:

- Finishing skill (goals relative to xG)
- High-danger shot share and finishing rate
- xG outperformance, primary assist share, on-target rate
- Powerplay production: PP ice time %, PP points/60, PP xG/60

Shot danger zones are weighted empirically: **high-danger 11.2×**, **medium-danger 4.1×**, **low-danger 1.0×**.

2. Career History Features

Computed strictly from prior seasons to prevent data leakage:

- Previous-season stats, 3-year rolling means, career means, career peak values
- Linear trajectory slopes (3-year and full-career) for points, goals, and game score
- Current % of career peak — included as a ceiling signal

3. Age Features

Included when age data is available:

- Age, age-squared (captures non-linear career arc)
- Interaction terms: age × shot attempts, age × finishing skill, age × high-danger share

These interaction terms allow the model to learn that identical skill levels carry different predictive weight at 25 vs. 33.

4. Team Context Features

Aggregated at team-position-season level. Captures system deployment:

- Median ice time/game, average adjusted xG/60, high-danger shot share
- Primary assist rate, on-target rate
- Linemate quality: line-level Corsi%, xG%, high-danger xG/60

At prediction time, these features are swapped out for each of the 32 NHL teams to simulate cross-team performance.

Two Model Versions

	Team Fit Model	Next Season Model
Training target	Current-season features → current-season targets	Current-season features → following-season targets
Core question	Given this player's skill profile today, how would they perform over the season?	What does this player's skill profile look like next season and what output should we expect?
Additional features	—	YoY deltas (points, goals, game score); GP/82; career year number
Holdout rule	None	Most recent season per player excluded (no future target exists)

Training Process

- **Cross-validation:** 3-fold CV used to generate performance metrics (MAE, RMSE, elite-segment MAE for top 10th percentile).
- **Production retraining:** Final model retrained on full dataset after CV evaluation.
- **Eligibility filter:** Players with <20 GP or <300 minutes of ice time excluded from training.
- **Elite upweighting:** Top-10th-percentile players (by output) weighted 3× during training to allocate additional model capacity to high-end prediction accuracy.

Metrics: MAE = Mean Absolute Error | RMSE = Root Mean Squared Error | xG = Expected Goals | GP = Games Played | YoY = Year-over-Year | PP = Power Play | Corsi = Shot attempt differential proxy